

SCIENCE MASTER X

Complete Academic Study Notes • Class 10 Science

CHAPTER 08

Heredity

(Full Chapter)

Accumulation of Variations • Mendel's Experiments • Monohybrid & Dihybrid Crosses • Laws of Inheritance • Sex Determination in Humans

Target Level: CBSE Class 10

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Chapter Overview & Basic Concepts

Heredity refers to the transmission of genetically based characters (traits) from parents to offspring across successive generations.

- **Genetics:** The branch of biology that deals with the study of heredity and variations.
- **Variations:** Differences observed in traits among individuals of the same species.
 - **Asexual Reproduction:** Generates minimal variation (due to minor DNA copying inaccuracies).
 - **Sexual Reproduction:** Generates maximum variation (due to combining parental gametes).
- **Genotype:** The underlying genetic constitution of an organism (e.g., TT, Tt, or tt).
- **Phenotype:** The observable physical expression of a trait (e.g., Tall or Dwarf).

NCERT High-Yield Insight: Inherited traits are expressed through cellular DNA. A section of DNA that provides information for synthesizing a specific protein is called a **Gene**. Genes control traits by regulating enzyme production (e.g., efficient plant hormone synthesis results in a tall plant).

SECTION 1: ACCUMULATION OF VARIATIONS & CONTRASTING TRAITS

Variations created in one generation are passed on to the next, accumulating over time to grant adaptive advantages in changing environments.

1.1 Why Mendel Chose Garden Pea (*Pisum sativum*)

- Availability of clear, contrasting visible traits.
- Short life cycle making multiple generations fast to study.
- Produces a large number of seeds in a single generation.
- Naturally self-pollinating, but easy to cross-pollinate artificially.

1.2 Key Terminology

Term	Definition & Concept
Alleles	Alternative forms of a gene occupying the same locus on homologous chromosomes (e.g., 'T' and 't').
Dominant Allele	An allele that expresses itself even in the presence of a contrasting allele (e.g., 'T' in Tt).
Recessive Allele	An allele that expresses itself only when present in identical pairs (e.g., 't' in tt).
Homozygous	Organism having identical alleles for a specific gene pair (e.g., TT or tt).
Heterozygous	Organism having different alleles for a specific gene pair (e.g., Tt).

SECTION 2: MENDEL'S LAWS & MONOHYBRID CROSS

A cross involving the inheritance of a single pair of contrasting characters (e.g., Plant Height: Tall vs. Dwarf) is called a **Monohybrid Cross**.

2.1 Monohybrid Cross Representation

- Parents (P₁):** Pure Tall (TT) × Pure Dwarf (tt)
- Gametes:** (T) and (t)
- F₁ Generation:** All Tt (Phenotypically 100% Tall due to dominance).
- F₂ Generation (Selfing F₁ × F₁):** Tt × Tt

P₁: TT × tt → Gametes: (T), (t) → F₁: Tt (100% Tall)

F₂ Punnett Square: (T, t) × (T, t) → TT (Tall), Tt (Tall), Tt (Tall), tt (Dwarf)

Diagram: Monohybrid Cross Punnett Square (F₂ Generation)

	T	t
T	TT (Tall)	Tt (Tall)
t	Tt (Tall)	tt (Dwarf)

2.2 Monohybrid Cross Ratios

Ratio Type	Observed Ratio	Breakdown
Phenotypic Ratio (F ₂)	3 : 1	3 Tall : 1 Dwarf
Genotypic Ratio (F ₂)	1 : 2 : 1	1 TT : 2 Tt : 1 tt

2.3 Derived Principles

- Law of Dominance:** Out of two contrasting alleles, only one expresses itself in F₁ (Dominant), while the other remains hidden (Recessive).
- Law of Segregation:** Allele pairs separate during gamete formation, so that each gamete carries only one allele for a trait.

SECTION 3: DIHYBRID CROSS & INDEPENDENT ASSORTMENT

A cross studying the inheritance of two pairs of contrasting characters simultaneously (e.g., Seed Shape: Round/Wrinkled & Seed Color: Yellow/Green) is called a **Dihybrid Cross**.

3.1 Dihybrid Cross Setup

- **Parents (P₁):** Round Yellow (RRYY) × Wrinkled Green (rryy)
- **Gametes:** (RY) and (ry)
- **F₁ Generation:** All RrYy (Phenotypically Round & Yellow).
- **F₂ Generation (F₁ × F₁):** RrYy × RrYy produces 16 combinations.

3.2 Dihybrid Ratios & Law

Phenotype Category (F ₂)	Genotypic Forms	Ratio Count
Round, Yellow (Both Dominant)	R-Y-	9
Round, Green (Dominant/Recessive)	R-yy	3
Wrinkled, Yellow (Recessive/Dominant)	rrY-	3
Wrinkled, Green (Both Recessive)	rryy	1

Dihybrid F₂ Phenotypic Ratio = 9 : 3 : 3 : 1

NCERT High-Yield Insight: The 9:3:3:1 phenotypic ratio proves the **Law of Independent Assortment**. It demonstrates that inheritance of seed shape is completely independent of the inheritance of seed color during gamete formation.

SECTION 4: SEX DETERMINATION IN HUMANS

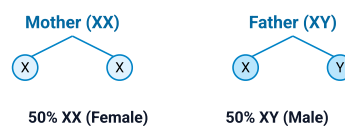
Human body cells contain **23 pairs (46)** of chromosomes in their nucleus.

- **Autosomes:** 22 pairs (44 chromosomes) responsible for general somatic traits.
- **Sex Chromosomes:** 1 pair (2 chromosomes) responsible for determining sex.
 - **Females:** Perfect matching pair → **XX** (Homogametic).
 - **Males:** Unmatched pair → **XY** (Heterogametic: normal X, short Y).

4.1 Sex Determination Mechanism

- All female gametes (eggs) carry one **X chromosome**.
- Male gametes (sperms) are 50% **X-carrying** and 50% **Y-carrying**.
- Sex of the child is strictly determined by the father's fertilizing sperm cell:
 - Egg (X) + Sperm (X) → **XX (Female Child)**
 - Egg (X) + Sperm (Y) → **XY (Male Child)**

📌 Diagram: Sex Determination Chart in Humans



? Important Practice Questions

Q1: A tall pea plant (TT) is crossed with a dwarf pea plant (tt). What will be the phenotype and genotype of the F₁ and F₂ generations?

Ans: F₁ Generation: All Tall (Genotype Tt).

F₂ Generation: Phenotype ratio = 3 Tall : 1 Dwarf; Genotype ratio = 1 TT : 2 Tt : 1 tt.

Q2: Why are small variations crucial for the survival of a species?

Ans: Variations provide adaptability to species during sudden drastic environmental changes (e.g., heat-resistant bacteria surviving a sudden increase in temperature), preventing complete extinction.

Q3: How do genes control the expression of traits in an organism?

Ans: A gene is a segment of DNA that codes for a specific protein/enzyme. The produced enzyme regulates metabolic pathways that lead directly to the expression of a particular physical trait (e.g., plant height).

Q4: "The sex of a child is determined by the father, not the mother." Justify this statement.

Ans: Mothers carry two identical X chromosomes (XX) and produce eggs carrying only X chromosomes. Fathers carry XY sex chromosomes and produce sperms carrying either X or Y (50% each). The child's sex depends on whether an X or Y sperm fertilizes the egg.

